

Organizational learning culture—the missing link between business process change and organizational performance

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Abstract

The aim of this paper is to present and test a model of organizational performance improvement based on the impact of organizational learning culture. The concept of organizational learning culture (OLC) is proposed and defined as a set of norms and values about the functioning of an organization. They should support systematic, in-depth approaches aimed at achieving higher-level organizational learning. The elements of an organizational learning process that we use are information acquisition, information interpretation, and behavioral and cognitive changes. Within the competing values framework OLC covers some aspects of all four different types of cultures: group, developmental, hierarchical, and rational. We use data from 203 Slovenian companies employing more than 50 people. The impact of OLC on organizational performance is empirically tested via structural equation modeling (SEM). The results show that OLC has a positive direct impact on all three aspects of non-financial performance included in the model: performance from the employee, customer, and supplier perspectives. The effect of organizational learning culture on financial performance is still positive, but indirect (through non-financial performance from the employee perspective).

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1. Introduction

Business and technological changes are threatening organizational sustainability and modern management faces many challenges (Drucker, 1999). Organizations are continually under competitive pressures and forced to re-evaluate their business models and underlying business processes. Operations management thus focuses on the careful management of internal processes along with processes in the supply chain, particularly by

improving their efficiency and effectiveness, which are today needed more than ever. Yet, despite this, not enough attention has been paid to this topic (Hammer, 2004).

The extensive literature on business process change (e.g. Davenport, 1993; Hammer and Champy, 1993; McCormack and Johnson, 2000; Burlton, 2001; Harmon, 2003) suggests that organizations can enhance their overall performance by adopting a process view of business. However, what is too often neglected is that most problems regarding business process management are not technical but arise from an inappropriate organizational culture that may impede innovations being implemented

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and superior performance being achieved (see e.g., Terziovski et al., 2003; Hammer, 2004).

The basic idea behind this paper is that organizational culture is very important when trying to improve organizational performance by business process change. The paper addresses organizational learning culture, which is proposed and defined as a set of norms and values about the functioning of an organization. It is a combination of different culture types within the competing values framework (Denison and Spreitzer, 1991; McDermott, and Stock, 1999). The purpose of the paper is to present and test a model of organizational performance improvement. Hence, the focus of this study is on the impact organizational learning culture has on organizational performance.

The outline of the paper is as follows: Section 2 reviews the relevant literature in order to demonstrate our specific contributions. Section 3 conceptualizes the research model leading to the development of suitable hypotheses. Section 4 aims to present a methodological framework for the study, while Section 5 provides results of data analysis. Section 6 concludes with a summary of the main findings, discusses them from theoretical and practical standpoints, and outlines directions for future research together with the limitations of the study.

2. Literature review

2.1. Business process change and organizational culture

Business process change can take many forms, ranging from continuous improvement with total quality management (TQM) to radical business process reengineering (BPR) (Harmon, 2003). However, the expected results have not been achieved in many cases. During the 1990s many firms undertook BPR projects and, despite its promising scheme, the success rate was well below the desired one. Different studies report different failure rates of the BPR project ranging from 60% to an almost 80% failure rate (Holland and Kumar, 1995; Smith, 2003; Terziovski et al., 2003). The outcomes of these studies fuelled new ones in order to identify the reasons for the failures.

The idea of determining the critical success factors (CSF) of business process change emerged as a response to the low success rates of BPR projects. One characteristic common to many CSF

frameworks is the notion that organizational culture is key to any change initiative. Altering the way people perceive changes and react to them plays an important role in such efforts (Grover et al., 1995; Marchand and Stanford, 1995; Wells, 2000). Culture has been shown to have a significant impact on how the objectives of a BPR project are accomplished (Clemons et al., 1995; Guimaraes, 1997; Al-Mashari and Zairi, 1999; Terziovski et al., 2003). Similar relationships have been studied for TQM. Researchers have consistently argued that any company choosing to adopt TQM must redefine its culture to some extent (Lewis, 1996; Abraham et al., 1997; Pool, 2000; Kekäle et al., 2004) and that success in implementing TQM depends on the organizational culture (Laszlo, 1998; Prajogo and McDermott, 2005).

Many different people have used the word ‘culture’ to explain a variety of phenomena. As each one tends to adopt a slightly different perspective, there is no universally accepted definition (Rollinson and Broadfield, 2002). Ott (1989) identified over 70 different words or phrases used to define organizational culture. One of the first attempts was by Jacques (1952) who claimed that organizational culture is the customary and traditional way of doing things, which is shared to a greater or lesser degree by all members, and which the new members must learn and at least partially accept in order to be accepted for the firm’s services. Harrison (1972) focused more on culture itself rather than on its effects and defined it as ideologies, beliefs, and deep-set values that occur in all firms and are prescriptions for the ways in which people should work in these organizations. Peters and Waterman (1982) saw culture as a dominant and coherent set of shared values conveyed by symbolic means such as stories, myths, legends, slogans, anecdotes and fairy tales. Deal and Kennedy defined organizational culture as ‘the way things get done around here’ (Deal and Kennedy, 1982, p. 90).

Schein (1992) perceived organizational culture as a pattern of basic assumptions—invented, discovered or developed by a given group as it learns to cope with its problems of external adaptation and internal integration. Such a pattern has worked well enough to be considered valuable and, therefore, to be taught to new members as the correct way to perceive, think and feel in relation to those problems. Wiener claimed that ‘most researchers of organizational culture agree that shared values

are a key element in the definition of culture' (Wiener, 1988, p. 534).

Organizational culture has many dimensions and variations. The competing values framework (CVF) categorizes them in a two-dimensional space (Denison and Spreitzer, 1991); see Fig. 1. Each axis represents contrast orientations. The first dimension stands for flexibility versus control orientation. The second dimension describes the focus on activities occurring within or outside the organization. The combination of both dimensions defines four types of organizational culture: group, developmental, hierarchical, and rational. Group culture emphasizes flexibility and change and a focus on the internal organization. Developmental culture also emphasizes flexibility, but is externally focused. Rational culture is externally oriented, but focused on control. Hierarchical culture emphasizes stability; however, the focus is on the internal organization. Characteristics of all four types of cultures are represented in Fig. 1 and are further described in Denison and Spreitzer (1991), McDermott and Stock (1999) and Prajogo and McDermott (2005).

An important assumption of CVF is that each type of culture is an ideal type. The culture in an organization is a combination of different culture orientations, although usually one type is more dominant than the others. 'A high rating on one dimension, e.g. internal orientation, does not exclude high rating at the other end, e.g., external orientation' (McDermott and Stock, 1999, p. 525). Further, Denison and Spreitzer (1991) argued that overemphasizing any culture type may become dysfunctional and the strength of the quadrant may even become a weakness.

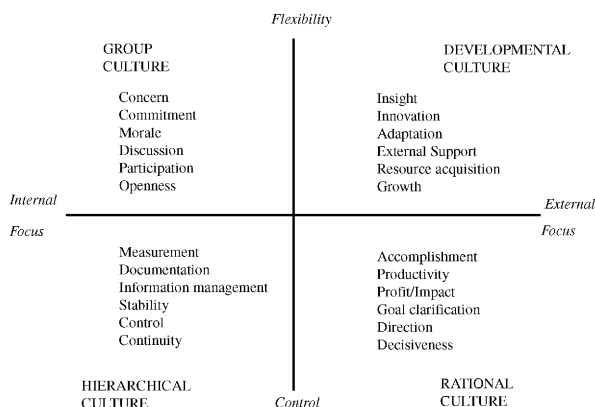


Fig. 1. The competing values framework (Denison and Spreitzer, 1991; McDermott, Stock, 1999).

While there is a consensus that organizational culture is critical in any change initiative, no such consensus exists as to what type of organizational culture best supports business transformation. Prajogo and McDermott (2005) studied the relationships between different types of cultures and certain TQM practices. The results show that different types of culture support different subsets of TQM practices. The findings also indicate that an organization can implement different, even opposite culture types, in harmony. This opened up the question of which combination of culture types is most appropriate for business process change. The organizational learning culture presented in the next subsection is a combination of all four cultural types as shown in Section 3. The purpose of this paper is to show that organizational learning culture leads to superior organizational performance.

2.2. Organizational learning culture

Organizational learning is a complex process that refers to the development of new knowledge and has the potential to change behavior (Huber, 1991; Slater and Narver, 1995). It is a time-honored process that involves changing individual and organizational behavior (Murray and Donegan, 2003). Firms that have developed a strong learning culture are good at creating, acquiring and transferring knowledge, as well as at modifying behavior to reflect new knowledge and insight (Huber, 1991; Garvin, 1993). Hence, organizations stressing organizational learning culture (OLC) must first acquire information, interpret it to fully understand its meaning and transform it into knowledge. At the same time, they must not forget the most important part—to implement behavioral and cognitive changes—in order to convert words into action.

Like organizational culture, organizational learning is also a very elusive concept due to the variety of perspectives that come under scrutiny in the academic literature. There have been numerous attempts to define organizational learning and its various aspects. Senge (1990) defined organizational learning as 'a continuous testing of experience and its transformation into knowledge available to whole organization and relevant to their mission' (Senge, 1990, p. 6), while Huber (1991) saw it as a combination of four processes: information acquisition, information distribution, information interpretation and organizational memory. Argyris and Schön (1996) were even less restrictive in their

definition by declaring that organizational learning emerges when organizations acquire information (knowledge, understandings, know-how, techniques and procedures) of any kind by any means.

Jones (2000) emphasizes the importance of organizational learning for organizational performance. He defines it as ‘a process through which managers try to increase organizational members’ capabilities in order to better understand and manage the organization and its environment’ (Jones, 2000, p. 472). Dimovski (1994) provided an overview of previous research and identified four varying perspectives on organizational learning. His model managed to merge informational, interpretational, strategic and behavioral approaches to organizational learning and defined it as a process of information acquisition, information interpretation and resulting behavioral and cognitive changes which should, in turn, have an impact on organizational performance.

A lack of empirical investigations into organizational culture on various aspects of organizational performance is still reported (e.g. Hammer, 2004). Only a few studies have tackled some aspects of this issue in recent years (Thompson and Kahnweiler, 2002; Egan et al., 2004; Kandemir and Hult, 2005). They all seem to rely on data from one functional type of employee (IT), one specific setting in which two companies collaborate (international joint venture), or insights gathered from several (four) companies, respectively. Their notion of organizational performance is narrowed down to specific items such as employee job satisfaction, the motivation to transfer learning (Egan et al., 2004), participation in decision-making (Thompson and Kahnweiler, 2002), or innovativeness and innovation capacity (Kandemir and Hult, 2005). Besides, they all observe organizational learning culture in economic entities in developed countries but fail to provide insights into the specifics of transitional economies.

Using the model developed in the paper we seek to show that an organizational learning culture can contribute to organizational performance and upgrade prior knowledge from several perspectives. First, we use the sample of all companies with more than 50 employees in a transitional economy. Second, we believe we have expanded the concept of organizational learning culture from the competency perspective alone to a notion that covers the process component while not disregarding the importance of linking learning opportunities with

organizational activities. Another specific contribution of our study is linked to organizational performance and the way we measure and evaluate it. Stemming from the stakeholder perspective (Freeman, 1984, 1994; Atkinson et al., 1997), we aim to encompass the perspectives of owners, employees, customers and suppliers by taking both financial and non-financial measures of organizational performance into account. While this approach is not novel in the scientific community, it has rarely been used in the context of transitional economies.

3. Conceptualization of the research model

3.1. Operationalization of the organizational learning culture construct

There have not been too many attempts to define and operationalize OLC in the past. For instance, Watkins and Marsick (1993, 2003) developed an analytical framework of the learning organization, which was used in the study by Egan et al. (2004) as a substitute for learning culture. Having done that, they seemed to neglect the fact that the OLC and learning process are implicitly intertwined. They also appeared to neglect that the integration of a four-stage process that includes information acquisition, information dissemination, information interpretation and organizational memory can provide a better understanding of the learning culture’s role in achieving superior organizational performance as shown by Kandemir and Hult (2005).

We define OLC as a set of norms and values about the functioning of an organization (Schein, 1992) that support systematic, in-depth approaches aimed at achieving higher-level i.e. double-loop (Argyris and Schön, 1996), deuterio (Schön, 1975), strategic (Bhattacharya, 1985) or generative (Wit-trock, 1974, 1990, 1992) organizational learning through phases of information acquisition, information interpretation and accompanying behavioral and cognitive changes (Huber, 1991; Garvin, 1993; Dimovski, 1994).

In order to explain the concept of OLC further, we use the competing values framework developed by Denison and Spreitzer (1991) and describe the main characteristics of OLC by placing them in the two dimensional space of CVF. The main characteristics of OLC are predominantly located within the flexibility orientation. Information gathering is

one of the key attributes of a learning culture. We group items (see Appendix A) within that construct into two scales: strategic information acquisition (SIA—IA4, IA6, IA10, IA12—for an explanation of the items see Appendix A) and operational and tactical information acquisition (OTIA—IA5, IA11, IA13). Strategic information is usually outwards-oriented, market-driven, focuses on the long-term future and is the basis for the firm's future direction. In that respect OLC possesses some of the characteristics of the developmental culture in CVF.

Operational and tactical information is more inward-oriented and short-term in nature. Day-to-day and short-term business decisions are based on this. Besides the firm's formal data sources (e.g. information systems) other modes of acquiring information are also important. For example, co-workers are often an invaluable source for this class of information. Operational information is usually acquired within the firm while tactical information comes from both within and outside the firm. Formal data sources are seldom sufficient so employees have to be innovative and flexible in their search for them. The acquisition of operational and tactical information requires group, developmental and some formal culture-type characteristics.

When data is available it has to be interpreted and placed within a specific environmental context in order to make sense and have value for the employees. The interpretation of information occurs through means of 'face-to-face' contacts (FACE—II2, II3), through electronic media such as the firm's intranet and forums (ELECTRO—II8, II9) and in more formal settings (e.g. a formal chain of command and written memos, letters etc.) (FORMAL—II5, II6, II7). In order for the first two modes to emerge, a firm's culture must have elements of a group culture as defined in CSV. Namely, there has to be an openness and willingness for discussion. Teamwork is also a good way for the exchange of ideas and information and for discussion. For a formal interpretation of information elements the Hierarchal culture type is more appropriate.

When information is available, understood and placed within a specific context, cognitive changes (COGNIT—BCC8, BCC9, BCC10, and BCC11) should emerge. Employees equipped with new information perceive things from a new angle or gain a deeper insight into the functioning of an organization and/or external environment. For this

to be true they have to be open-minded. Clearly, cognitive changes require flexible and adaptive employees. In this respect, OLC possesses elements of group and developmental culture.

The learning process is completed when employees change their behavior (BEHAV—BCC2, BCC4, BCC5, BCC7) based on information they received and the way they understand it. The changes in behavior are rational in their nature. The purpose of behavioral changes is to achieve goals, which can mean many different things—ranging from productivity increases to higher efficiency achievements and overall performance improvement. The rationality of behavioral changes coincides with the characteristics of the rational culture from CVF.

Organizational learning culture thus contains elements of all four ideal types of cultures defined in CVF (as evident from Fig. 2). In its essence, OLC is a flexible culture that acknowledges both internal and external environments. The flexibility is complemented with some elements of the Control dimension that provide the clarity, structure and formal reference framework needed for the firm's successful functioning.

3.2. Organizational performance

One cannot evaluate organizational performance without taking organizational goals into consideration. The modern business environment demands a multi-goal orientation. Profit theory (Cyert and

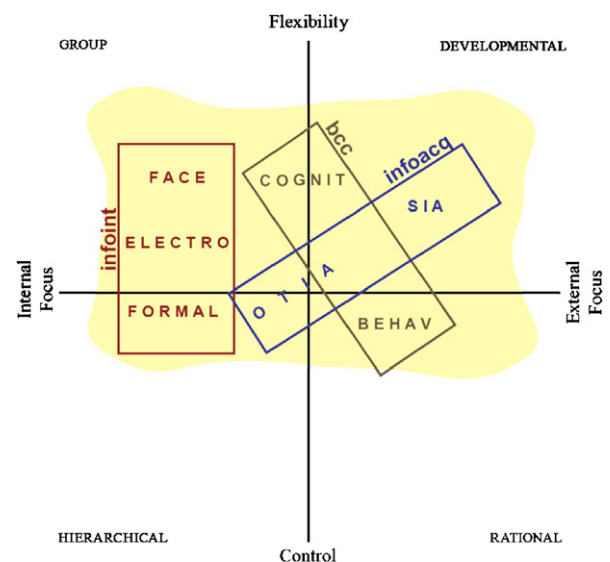


Fig. 2. The placement of organizational learning culture in the competing values framework.

March, 1963) is no longer a valid measure of organizational performance and neither are other approaches that only take the interests of shareholders (owners) of a company into account. Indeed, Agrell et al. state that ‘although a profit maximizing behavior may be induced by profit-sharing schemes, such contracts may easily lead to sub-optimal levels of organizational training, innovation, and knowledge transfer’ (Agrell et al., 2002, p. 1). Today’s business environment is characterized by the increasing importance and strength of various stakeholder groups.

It has become quite obvious that all stakeholders need to be taken into account when assessing a modern company’s performance. This is the main idea of Freeman’s Stakeholder theory (Freeman, 1984, 1994). The stakeholder view maintains that firms have stakeholders rather than just shareholders to account for. The view that the corporation has obligations only to its stockholders is replaced by the notion that there are other groups to whom the firm is also responsible. Groups with a ‘stake’ in the firm include shareholders, employees, customers, suppliers, lenders, the government and society (Berman et al., 1999; Harrison and Freeman, 1999; Hillman and Keim, 2001; Riahi-Belkaoui, 2003).

Behavioral theory (Cyert and March, 1963) recognized the company as a coalition of individuals or groups of individuals such as management, employees, customers, owners, government, etc. but did nothing to introduce this finding into organizational performance assessment. Emerging management paradigms are emphasizing a stakeholder perspective (Atkinson et al., 1997; Berman et al., 1999; Harrison and Freeman, 1999; Hillman and Keim, 2001; Sirgy, 2002; Riahi-Belkaoui, 2003; Tangem, 2004).

One important notion revealed in many studies is that building better relations with primary stakeholders like employees, customers and suppliers could lead to increased shareholder wealth. A sustainable organizational advantage may be built with tacit assets that derive from developing relationships with key stakeholders (Hillman and Keim, 2001). When studying the relationship between stakeholder management and a firm’s financial performance, Berman et al. (1999) found that fostering positive connections with key stakeholders (customers and employees) can help a firm’s profitability.

Due to the significance of various stakeholders, organizational performance should not be solely assessed by financial indicators. There are several

approaches (Tangem, 2004) to organizational performance measurement that encompass different stakeholders perspectives. The Balanced Scorecard (BSC) (Kaplan and Norton, 1992, 1993, 1996) is the most established and most commonly used (Neely, 2005), but by far not the only one. The multi-model performance framework (MMPF) model by Weerakoon (1996) is also very interesting and has four dimensions including employee motivation, market performance, productivity performance, and societal impact, and covers the satisfaction of various stakeholders such as customers, investors, employees, suppliers and society. A more recently developed conceptual framework is the performance prism, which suggests that a performance measurement system should be organized around five distinct but linked perspectives of performance (Tangem, 2004, p. 733).

3.3. *Research hypotheses and model*

Several authors have studied the impact of organizational culture(s) on various aspects of organizational performance (e.g. Denison et al., 2003; Deshpandé and Farley, 2004; Flamholtz and Kannan-Narasimhan, 2005; Yilmaz et al., 2005). Likewise, quite a few studies have emerged in recent times that have scrutinized the relationship between organizational learning process and organizational performance (Adler, 1990; Adler and Clark, 1991; Dimovski, 1994; Figueiredo, 2003; Lam, 1998; Pisano et al., 2001; Simonin, 1997; Sloan et al., 2002; Vits and Gelders, 2002). However, not many studies have merged these two fields and evaluated the impact of organizational learning culture on organizational performance.

Thompson and Kahnweiler (2002) base their work on four organizations (with 97 respondents) on the theoretical proposition that there is a relationship between employee participation in decision-making and organizational learning culture and limit their model to Schein’s theory of OLCs (1992). Unfortunately, their results do not offer strong support for their proposal. On the contrary, Egan et al. (2004) studied the impact of organizational learning culture on turnover intention via job satisfaction on a sample of 245 IT employees from thirteen US-based firms and found that higher-level organizational learning culture is associated with the IT employees’ job satisfaction and both directly and indirectly with lower turnover intention. Similarly, Kandemir and Hult (2005) explore

organizational learning culture in international joint ventures and link it to one specific aspect of performance—innovations. They propose that organizational learning culture has a positive effect on its cultural innovativeness and innovation capacity.

In their work on organizational culture(s) and TQM practices [Prajogo and McDermott \(2005\)](#) find support for the pluralist view, which ‘suggests the existence of multidimensional cultures’ (p. 1107). It is in this way that we understand organizational learning culture as a combination of values and norms that support group, developmental, and to some extent also hierarchical and rational culture (according to CVF). We start from the basic research question which deals with the question of the impact of organizational learning culture on organizational performance, and develop hypotheses accordingly.

Organizational learning process is a sequence of three phases: information acquisition, information interpretation, and behavioral and cognitive changes. Organizations with a strong organizational learning culture place high importance on the acquisition of operational, tactical and strategic information. Information can be regarded as a raw material for learning. In the next phase, this information needs to be transformed into meaning through the information interpretation phase. Firms that value the interpretation of information use face-to-face, formal and electronic channels. For learning to happen, information needs to be acquired, understood, and above all transformed into action ([Garvin, 1993](#); [Huber, 1991](#)). Both behavioral and cognitive changes in the functioning of organizations are needed for learning to be effective ([Murray and Donegan, 2003](#); [Garvin, 1993](#)). All three phases of the organizational learning process need to be assigned with a high level of importance in order to claim that an organization has a strong learning culture. On this basis we pose the first two hypotheses:

H₁: Ascribing greater importance to the acquisition of operational, tactical and strategic information (Infoacq) leads to the better interpretation of information (Infoint).

H₂: Assigning greater importance to formal, face-to-face and electronic channels of interpreting information (Infoint) leads to more action in terms of behavioral and cognitive changes (Bcc).

Another set of hypotheses should relate organizational learning culture to organizational perfor-

mance. Positive changes in the way people act (behavioral changes) and perceive their internal and external environments (cognitive changes) are expected to have a positive impact on organizational performance (e.g. [Egan et al., 2004](#); [Kandemir and Hult, 2005](#); [Thompson and Kahnweiler, 2002](#)). Specifically, changing actions and cognitive maps of members of an organization should lead to improved performance from the employee, customer, and supplier perspectives. Employees in organizations with a strong learning culture feel that their managers share information and that they are empowered to take the necessary steps to adapt to changes in internal and external business environments. Consequently, this should positively reflect on commitment, trust, preparedness to go an extra mile for the company (subjective measures) and the productivity of employees, work costs and absenteeism (objective measures of performance from the employee perspective). At the same time, a strong organizational learning culture should mean that an organization is better in dealing with its customers in terms of attracting new clients, retaining old (profitable) ones, and an improved reputation in the customers’ eyes. An organizational culture that values learning should also show benefits on the other side of the value chain—in terms of improved relationships with suppliers. If members of an organization have the necessary information, fully understand its meaning and opportunities and are able to convert it into action, this should mean that it can maintain better relationships with suppliers. In this context, hypotheses 3–5 are put forward:

H₃: Improved Bcc will have a positive impact on performance from the employee perspective (Emp).

H₄: Improved Bcc will have a positive impact on performance from the customer perspective (Cust).

H₅: Improved Bcc will have a positive impact on performance from the supplier perspective (Sup).

The next group of hypotheses will relate organizational learning culture to financial performance. There were some indications from the pilot study we conducted that organizational learning culture might have a direct impact on financial performance. While this speculation would be highly probable with some time lag, it is still worthwhile testing it in a cross-sectional model. We decided to

include this speculation in the model in the form of hypothesis 6.

H₆: Improved Bcc lead to improved financial performance (Fp) in terms of an increased return on assets and value added per employee.

More probably, organizational learning culture will indirectly affect financial performance. There are at least three possible ways through which organizational learning can have an indirect effect on financial performance. First, the effect is evident through performance from the employee perspective. A productive and committed workforce that has trust in management is rarely absent from work and which works at lower costs than the competition will indisputably lead to more profit and higher value added per employee. Second, OLC might have an indirect effect on financial performance through better relationships with customers. Both hypotheses (H₇ and H₈) are consistent with the research of Berman et al. (1999). Again, there might be a time-lag effect which would demand longitudinal instead of cross-sectional observations but we decided to test this hypothesis as well. Third, regarding the indirect impact of organizational learning culture on financial performance through better relationships with suppliers, the picture is not that clear. On one hand, stable and long-term relationships with

the main suppliers mean stability which is desirable for long-term growth. Companies involving suppliers in the process of research & development of their materials gain in many respects. On the other hand, some pressure on the suppliers and the ability to switch among them might also lead to at least short-term cost savings. Still, we decided to test for positive effects of all three forms of non-financial performance on Fp.

H₇: A higher performance from the Emp is conducive to improved Fp.

H₈: A higher performance from the Cust improves Fp.

H₉: A higher performance from the Sup leads to a better Fp.

In Fig. 3 we illustrate the conceptualized research model in which all the main constructs are shown together with the hypothesized relationships among them.

4. Research framework and methodology

4.1. Research instrument

In order to test our hypotheses, the research paradigm proposed by Koufteros (1999) was used. We started developing our questionnaire by

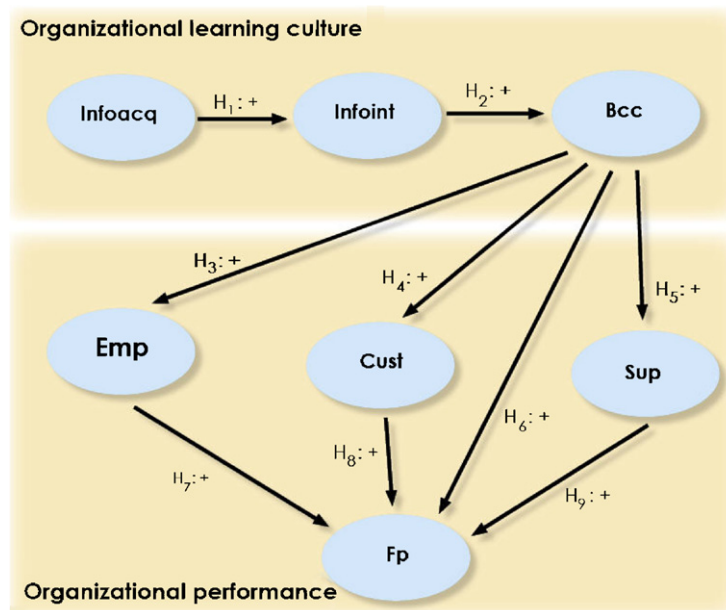


Fig. 3. The conceptualized research model.

building on the previous theoretical basis in order to ensure content validity. Pre-testing was conducted using a focus group involving six academics interested in the field and 10 semi-structured interviews with selected CEOs who were later also the subject of the study. Besides, a pilot study was conducted in 2003 on a sample of Slovenian companies with more than 100 employees. On that basis, a set of measurement items was expanded to include 38 items for organizational learning culture and 19 for various aspects of organizational performance (see Appendix A). We used a structured questionnaire with 5-point Likert scales for the organizational learning culture items and 5-point semantic differentials for those items measuring organizational performance.

4.2. The organizational learning culture measures

As discussed in the literature review, our understanding of organizational learning culture relates the process of organizational learning (Huber, 1991) to the competing values framework of organizational culture (Denison and Spreitzer, 1991; McDermott and Stock, 1999; Prajogo and McDermott, 2005). Hence, we understand the concept of organizational learning culture to be composed of three constructs: information acquisition (Infoacq), information interpretation (Infoint), and Bcc. Seven composite scores of items are used for their formation.

Members of an organization acquire short- to medium-term OTIA as well as long-term strategic information (SIA). Their sources come from within and outside of the company. In order to understand the meaning of the information acquired they need to employ various means and ways for interpreting the information: (1) face-to-face interpretation (FACE); (2) formal interpretation (FORMAL) using a formal chain of command reporting, written memos, notes, etc.; and (3) information interpretation using electronic means of communication such as intranet and e-forums (ELECTRO). For learning to happen, the acquired and interpreted information needs to lead to action in the form of behavioral (BEHAV) and cognitive (COGNIT) changes. In our understanding, organizational learning culture covers (to various extents) all four types of cultures according to the competing values framework (Denison and Spreitzer, 1991; McDermott and Stock, 1999; Prajogo and McDermott, 2005). It mostly involves flexibility-oriented cultures

(group and developmental culture), while some elements of hierarchical and rational culture are also present.

4.3. The organizational performance measures

Organizational performance is far wider concept than just profit or some other financial performance measures. As discussed earlier in this paper, beside shareholders other stakeholder groups also need to be taken into account when considering organizational performance. In that manner, we used four constructs of organizational performance: Fp; performance from the Emp; performance from the Cust; and performance from the Sup. Several perceptual measures of performance (relative to competition) were used for each of them.

Construct Fp is measured using ROA and VAEMP. The construct Emp is measured using both 'softer', subjective measures of performance (EMPSUB) as well as 'harder', more objective measures of performance (EMPOBJ). The former involves items such as trust in leadership, commitment and the preparedness to go 'the extra mile', while the latter combines employee productivity, costs of work, and absenteeism. The quality of relationships with suppliers (SUPPLIER) was the only item to measure the Sup construct. In order to measure the Cust construct we used four items aimed at establishing the extent of an increase/decrease in customer complaints (CUST1), the speed of dealing with customer complaints (CUST2), success in retaining clients and attracting new ones (CUST3), and the company's reputation in the eyes of customers (CUST4).

4.4. Data collection and sample characteristics

In autumn 2005, empirical data were collected through a survey of 1267 Slovenian companies that had more than 50 employees. Questionnaires were addressed to CEOs or senior managers estimated as having adequate knowledge of the organizational culture and performance within their companies. A total of 203 managers responded while, at the same time, 68 questionnaires were returned to the researchers with 'return to sender' (RTS) messages, indicating that the addresses were no longer valid or the companies had ceased to exist. We followed the approach of Prajogo and McDermott (2005) and discounted the number of RTS mails so the final response rate was 16.9%.

We originally aimed at an audience of top managers to ensure a strategic and to some degree even an interdisciplinary perspective on the company in question. However, there was some discrepancy between the desired and actual structures of respondents as shown in Table 1.

Based on the criterion of the average number of employees in 2004, 36.9% of the companies had between 50 and 249 employees, followed by 35.4% with 250–499 employees, while 18.1% had 500–999% and 9.6% of the companies had 1000 and more employees. Table 2 demonstrates the industry structure of the companies under investigation. Given that non-profit organizations were excluded from the study, the sample is an adequate representation of the population of Slovenian companies that have more than 50 employees.

Table 1
Structure of respondents by their function within the company

Respondent's function	Share (in%)
CEO or managing director of company	35.5
Data not available	24.1
Other: lower levels in hierarchy	16.3
Director of HRM/personnel	8.4
Marketing or public relations director	4.4
Director of finance and/or accounting or controlling	3.9
Production director	3.0
Director of quality	2.0
Director of informatics, CIO	1.5
Other	1.0
Total	100.0

Table 2
Structure of respondents by industry type

Industry (SIC)	Share (in%)
A Agriculture, hunting and forestry	2.5
B Fishing	0
C Mining and quarrying	1.0
D Manufacturing	46.8
E Electricity, gas and water supply	1.5
F Construction	10.3
G Wholesale and retail trade	10.3
H Hotels and restaurants	3.4
I Transport, storage and communication	6.4
J Financial intermediation	3.4
K Real estate, renting and business activities	1.0
O Other community, social and personal service activities	9.9
Not given	3.4

4.5. Research methods

For the purpose of validating the measurement instrument and modeling the structural relationships among the various constructs of organizational learning culture and organizational performance we used a combined exploratory–confirmatory approach. We followed the paradigm applied by Koufteros (1999). First, data were subject to exploratory factor analysis (see Appendix B) which may provide some initial insight but does not provide an explicit test of unidimensionality (Gerbing and Anderson, 1988; Segars and Grover, 1993). In addition, we applied confirmatory factor analysis (CFA) using the LISREL 8.54 software package. We examined convergent validity and unidimensionality by examining the loading paths of all items, which should be statistically significant and exceed 0.50 (Hair et al., 1998; Prajogo and McDermott, 2005). In the iterative process of purifying the scales several items (measurement variables) were excluded from further analysis. In the final version of the model, 34 of 57 items were used to measure 15 constructs and 7s-order factors. Simple second-order models for each of the second-order factors Infoacq, Infoint, Bcc, and Emp were run prior to combining the constructs OTIA, SIA, ELECTRO, FACE, FORMAL, BEHAV, COGNIT, EMPSUB, and EMPOBJ into the aggregates of items involved.

Discriminant validity testing is aimed at examining the extent to which constructs are distinct from each other (Prajogo and McDermott, 2005). We evaluated discriminant validity using a method whereby models are constructed for all possible pairs of latent variables within each instrument: (1) for a correlation between latent variables fixed at 1.0; and (2) for a correlation free to assume any value (Venkatraman, 1989; Koufteros, 1999). Subsequently, the χ^2 difference test is employed. A significantly lower χ^2 value would indicate that the traits are not perfectly correlated and that discriminant validity can be inferred (Bagozzi and Philips, 1982; Anderson, 1987; Bagozzi et al., 1991; Koufteros, 1999, p. 482). Altogether, 66 tests of discriminant validity were performed.

A composite reliability index (CRI) and average variance extracted (AVE) were calculated to test for composite (construct) reliability (Fornell and Larcker, 1981). Composite reliability assumes that a set of latent construct indicators is consistent in the measurement. There is no generally acceptable standard for adequate values of CRI. Koufteros

(1999) suggested values above 0.80, while Diamantopoulos and Siguaw (2000) were satisfied with 0.60. AVE is similar to CRI with the one exception that standardized loadings are squared before summing them (Hair et al., 1998; Koufteros, 1999). The cut-off value most often used for AVE is 0.50 (Bagozzi and Yi, 1988; Hair et al., 1998), while there are also cases where a milder restriction of 0.40 was employed (Diamantopoulos and Siguaw, 2000).

For the purposes of evaluating fit at the global level a plethora of fit indices exists. Research evidence supports the need to use more than one index (Breckler, 1990; Bollen and Long, 1993; Tanaka, 1993; Coenders et al., 2003). χ^2 per degrees of freedom, comparative fit index (CFI) and non-normed fit index (NNFI; also named the Tucker–Lewis fit index: TLI) are used most often to assess model fit (Koufteros, 1999). The ratio χ^2 per degrees of freedom should not exceed 2, while models exhibiting CFI and NNFI indices greater than 0.90 have an adequate fit. Some researchers (Coenders et al., 2003) even suggest a cut-off value of 0.95. We also have to stress the fact that the multivariate normality test showed the non-normal distribution of data for which a Satorra–Bentler χ^2 was used.

Finally, we used SEM to test the structural relationships among constructs. We decided to use SEM for similar reasons as Prajogo and McDermott (2005): (1) to allow for the modeling of both observed and latent variables; and (2) to test several structural relationships simultaneously. Specifically, the maximum likelihood (ML) method was used to estimate the parameter values. Even though several methods can be used for this purpose, ML is the one most frequently used and has the advantage of being statistically efficient and at the same time specification-error sensitive because it demands only complete data and does not allow for missing values. All methods will, however, lead to similar parameter estimates where the sample is large enough and the model is not misspecified (Jöreskog and Sörbom, 1993). Overall coefficients of determination (R^2) are reported for each endogenous variable to explain the amount of variation in an endogenous variable explained by the proposed model.

5. Data analysis

5.1. Validity and reliability

The construct validity of each scale was assessed using CFA in order to establish how well the items

measured the corresponding scales. In Table 3 we report unstandardized and completely standardized factor loadings together with corresponding t -values for each item and scale involved in the final solution of the measurement model. Four more items were excluded from further analysis (following the 19 omitted after the preliminary exploratory factor analysis). All items and scales (except IA4 and FACE) exceed the threshold of 0.50 for convergent validity. To avoid data-driven modifications we did not exclude item reports from external experts (IA4) nor scale face-to-face interpretation (FACE) from further analysis despite the fact they did not reach the required threshold. The justification for this decision makes sense from a substantive point of view and will be explained in the next paragraph. In general, both unidimensionality and convergent validity were confirmed.

Analysis of second-order models for Infoacq, Infoint, Bcc, and Emp provided empirical justification for combining constructs OTIA, SIA, ELECTRO, FACE, FORMAL, BEHAV, COGNIT, EMPSUB, and EMPOBJ into aggregates. Fit indices for all four second-order models are satisfactory (all $\chi^2/\text{df} < 2$, all NNFI > 0.90 , and all CFI > 0.95). All measurement variables are statistically significant related to constructs ($p < 0.05$) while the standardized loadings range from 0.52 to 0.88.

The values of the CRI as well as AVE are presented in Table 4 for all scales and constructs in the final measurement model. While all CRI values easily exceed the milder threshold for composite reliability (0.60), most of them also exceed the stricter criterion (0.80). The same can be said for AVE where values for all latent variables and scales (but one) exceed 0.50. The only exception is the scale face-to-face information interpretation (FACE). Nevertheless, it was left in the model due to content considerations. It is known from social network research (Cross and Parker, 2004) that people most likely turn to other people when they search for an explanation of certain information. We have to be aware though that CEOs were respondents in our survey. One trait of Slovenian national culture regarding the supervisor–subordinate relationship is that we expect to have bosses who know everything. This anomaly related to the validity and reliability of FACE could well be a result of a fear of losing face by turning to a subordinate for help. Another issue might be related to trust which has to be present in such a relationship. We did not control for its effect since it is a

Table 3
Construct validity

Second-order factors	Constructs	Measurement variables (items–final)	Unstandardized factor loading		Completely standardized factor loadings		t/values	
Infoacq	OTIA	IA5	1.00	1.00	0.65	0.56	— ^a	— ^a
		IA11		0.92		0.67		5.95
		IA13		0.73		0.55		4.99
	SIA	IA4	0.99	1.00	0.67	0.42	6.80	— ^a
		IA6		1.20		0.51		4.20
		IA10		1.20		0.50		4.36
		IA12		1.94		0.74		4.24
Infoint	ELECTRO	II8	1.00	1.00	0.66	0.81	— ^a	— ^a
		II9		0.76		0.66		5.05
	FACE	II2	0.57	1.00	0.45	0.76	4.76	— ^a
		II3		1.14		0.78		4.81
	FORMAL	II5	0.88	1.00	0.51	0.66	5.15	— ^a
		II6		1.34		0.78		6.54
		II7		0.90		0.55		5.63
Bcc	BEHAV	BCC2	1.00	1.00	0.74	0.61	— ^a	— ^a
		BCC4		1.31		0.72		7.24
		BCC5		1.12		0.63		6.73
		BCC7		1.16		0.64		6.54
	COGNIT	BCC8	1.10	1.00	0.69	0.89	7.75	— ^a
		BCC9		0.94		0.88		18.70
		BCC10		0.76		0.68		9.24
Emp	EMPSUB	BCC13		0.60		0.54		6.84
		EMP3	1.00	1.00	0.76	0.74	— ^a	— ^a
		EMP6		1.16		0.84		11.71
		EMP7		1.18		0.88		10.65
	EMPOBJ	EMP2	0.75	1.00	0.68	0.81	8.58	— ^a
		EMP8		0.60		0.52		5.01
		EMP9		0.68		0.52		5.08
Cust	CUST2	CUST2	1.00		0.71		— ^a	
	CUST3	CUST3	1.24		0.75		9.23	
	CUST4	CUST4	1.39		0.87		10.63	
Sup	SUPPLIER	SUPPLIER	1.00		1.00		— ^a	
Fp	ROA	ROA	1.00		0.76		— ^a	
	VAEMP	VAEMP	1.27		0.96		9.21	

Fit indices: $\chi^2 = 97.08$ ($p = 0.0178$), $df = 70$, $\chi^2/df = 1.39$, NNFI = 0.94, CFI = 0.96.^aIndicates a parameter fixed at 1.00 in the original solution.

very individual phenomenon and a different type of study would be needed.

Discriminant validity tests are presented in Table 5. Each value in the table stands for the χ^2 difference between models with a fixed correlation between constructs and models where the correlation is freely estimated. Given that all of the values exceed the critical χ^2 value of 10.83 with 1 degree of freedom ($p < 0.001$), discriminant validity can be inferred for all pairs of constructs.

5.2. The relationship between organizational learning culture and organizational performance

As noted earlier, this research aimed to test the structural relationship between organizational learning culture and organizational performance. Organizational learning culture is a culture in which the process of Infoacq, Infoint, and Bcc is highly valued. When information is acquired it needs to be interpreted and converted into

action in order to be able to say that organizational learning has happened. This is the reasoning behind the sequential structuring of elements of an organizational learning culture. Organizational culture aiming at optimizing this process can be considered an organizational learning culture. It covers both flexibility-oriented cultures (group and developmental) as well as hierarchical and rational cultures—in line with the competing values framework (Denison and Spreitzer, 1991; McDermott and Stock, 1999). Organizational

performance was evaluated from the stakeholder perspective. Construct performance from the Emp, performance from the Cust and performance from the Sup were considered antecedents to Fp.

The results of fitting the structural model to the data show that the model had a good fit as pointed out by $\chi^2/df = 1.26$, NNFI = 0.96, CFI = 0.97. In Fig. 4 the path diagram of our model is presented. When examining the structural part of the diagram, standardized values of path coefficients are presented with *t*-values in brackets. Overall coefficients of determination (R^2) are also presented for each one of the endogenous constructs. As mentioned in the research methods subsection, a mix of exploratory and confirmatory approaches was used for generating the model. The final goal was to set up a model that makes both theoretical sense and has a reasonable correspondence to the data (Jöreskog, 1993; Bollen and Long, 1993; McCallum, 1995; Prajogo and McDermott, 2005, p. 116). From a substantive point of view, nine relationships among constructs of interest were hypothesized. In the final version of the model, six out of these nine relationships were found to be statistically significant at $p < 0.001$.

Valuing the acquisition of operational, tactical and strategic information leads to a better understanding and interpretation of the acquired information. The effect is very strong (standardized value = 0.94), statistically significant ($t = 6.29$) and positive. Placing a high level of importance on various channels of information interpretation (face-to-face, electronic, formal) leads to greater action in terms of behavioral and cognitive changes

Table 4
Construct reliability

Latent variable	Aggregates of items	Number of items (final)	CRI		AVE	
Infoacq	OTIA	3	0.72	0.80	0.56	0.49
	SIA	4		0.72		0.40
Infoint	ELECTRO	2	0.66	0.80	0.39	0.67
	FACE	2		0.84		0.72
	FORMAL	3		0.80		0.57
Bcc	BEHAV	4	0.78	0.83	0.64	0.55
	COGNIT	4		0.91		0.74
Emp	EMPSUB	3	0.89	0.92	0.65	0.79
	EMPOBJ	3		0.75		0.51
Cust	CUST2	1	0.89		0.73	
	CUST3	1				
	CUST4	1				
Sup	SUPPLIER	1	1.00		1.00	
Fp	ROA	1	0.91		0.84	
	VAEMP	1				

Table 5
Discriminant validity

Scale/construct	1	2	3	4	5	6	7	8	9	10	11	12
(1) OTIA	—											
(2) SIA	37.12	—										
(3) ELECTRO	23.72	39.79	—									
(4) FACE	49.90	48.29	64.07	—								
(5) FORMAL	37.54	45.67	34.02	47.26	—							
(6) BEJAV	56.57	76.54	49.2]	68.77	70.04	—						
(7) COGNIT	41.26	60.44	40.43	59.83	63.61	48.79	—					
(8) EMPSUB	49.05	64.51	51.76	67.14	60.90	63.38	42.62	—				
(9) EMPOBJ	39.82	60.95	39.85	49.99	53.78	56.90	44.34	38.87	—			
(10) Cust	45.98	147.06	68.11	82.52	94.77	68.88	44.73	43.05	37.88	—		
(11) Sup	66.71	55.62	320.42	277.04	74.45	47.67	53.15	45.89	37.90	54.53	—	
(12) Fp	57.36	66.30	47.65	75.82	79.28	50.54	44.77	36.37	45.27	46.46	167.02	—

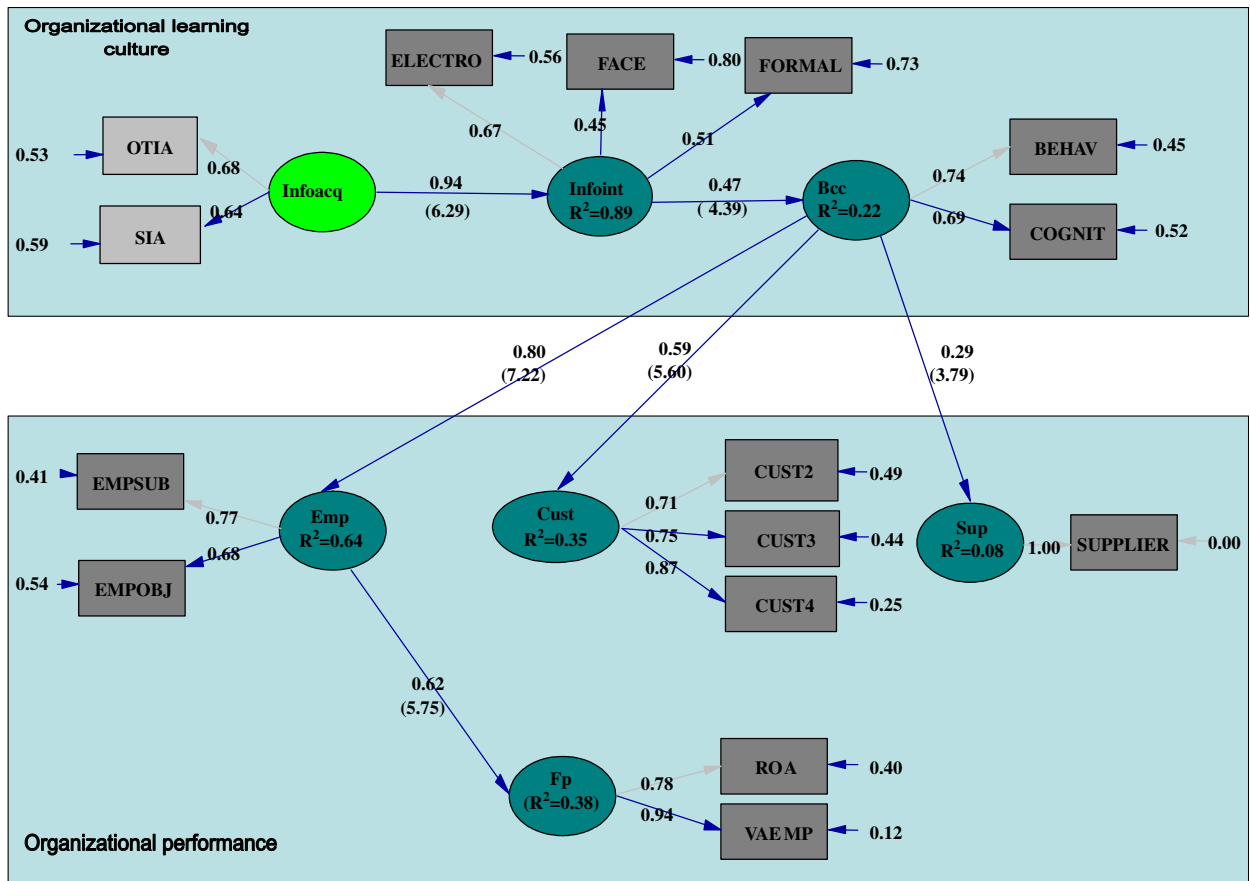


Fig. 4. The model of relationships among organizational learning culture and organizational performance constructs.

(effect = 0.47, $t = 4.39$), meaning that more learning has actually occurred.

The behavioral and cognitive changes construct has a statistically significant, positive and strong effect on all three aspects of non-financial performance included in the model. This effect is strongest on performance from the employee perspective (0.80, $t = 7.22$), while it is also strong on performance from the customer perspective (0.59, $t = 5.60$). The effect of behavioral and cognitive changes on performance from the supplier perspective is moderate in size (0.29) but still positive and statistically significant ($t = 3.79$). The only construct in the model that exhibited a direct impact on financial performance was performance from the employee perspective. This effect is statistically significant ($t = 5.75$), positive and strong (standardized value = 0.62).

Surprisingly, improved performance from the cust and sup did not exhibit improved performance

in financial terms. Also contrary to our expectations, behavioral and cognitive changes demonstrated no statistically significant effect on Fp. Obviously, positive financial effects of attaining an OLC emerge through improved performance from the Emp as is evident from the decomposition of effects seen in Table 6. Even though not strong, all elements of OLC (Infoacq, Infoint, and Bcc) have a statistically significant indirect effect on Fp.

6. Discussion and implications

Hypotheses 1 and 2 show that organizational learning is indeed a process in which information as a raw material is transformed into action. Organizations that value systematic approaches to organizational learning thus stress the importance of acquiring all types of information (operational, tactical and strategic) from both internal and external sources. The better a certain firms is at

Table 6
Decomposition of effects

Path	Unstandardized coefficients (<i>t</i> -values)			Standardized coefficients		
	Total effects	Direct effects	Indirect effects	Total effects	Direct effects	Indirect effects
Infoacq → Infoint	0.64 (6.29**)	0.64 (6.29**)	—	0.94	0.94	—
Infoacq → Bcc	0.41 (4.25**)	—	0.41 (4.25**)	0.44	—	0.44
Infoacq → Emp	0.35 (3.71**)	—	0.35 (3.71**)	0.36	—	0.36
Infoacq → Cust	0.08 (3.65**)	—	0.08 (3.65**)	0.26	—	0.26
Infoacq → Sup	0.06 (2.87*)	—	0.06 (2.87*)	0.13	—	0.13
Infoacq → Fp	0.09 (3.42**)	—	0.09 (3.42**)	0.22	—	0.22
Infoint → Bcc	0.64 (4.39**)	0.64 (4.39**)	—	0.47	—	—
Infoint → Emp	0.55 (3.80**)	—	0.55 (3.80**)	0.38	—	0.38
Infoint → Cust	0.12 (3.77**)	—	0.12 (3.77**)	0.28	—	0.28
Infoint → Sup	0.10 (2.80*)	—	0.10 (2.80*)	0.14	—	0.14
Infoint → Fp	0.14 (3.50**)	—	0.14 (3.50**)	0.23	—	0.23
Bcc → Emp	0.87 (7.22**)	0.87 (7.22**)	—	0.80	0.80	—
Bcc → Cust	0.20 (5.60**)	0.20 (5.60**)	—	0.59	0.59	—
Bcc → Sup	0.16 (3.79**)	0.16 (3.79**)	—	0.29	0.29	—
Bcc → Fp	0.23 (5.15**)	—	0.23 (5.15**)	0.49	—	0.49
Emp → Fp	0.26 (5.75**)	0.26 (5.75**)	—	0.62	0.62	—

* $p < 0.01$; ** $p < 0.001$.

acquiring information the more understanding it can get from it. In other words, information acquisition positively effects information interpretation, which is nothing other than the ability to recognize entrepreneurial opportunities. Behavioral and cognitive changes mean transforming words into actions and seizing these opportunities, which wraps up the organizational learning cycle. Firms that attribute a high level of importance to the elements of this process integrate them into their set of norms and values and may be considered to have an OLC.

The paper has shown that an OLC does have an impact on organizational performance. Specifically, we found that an OLC has a direct impact on all three studied elements of non-financial performance (Hypotheses 3–5). This effect is strongest when performance from the employee perspective is concerned. The effect of Fp was not direct (Hypothesis 6), but indirect and evident through employee performance (Hypothesis 7). Hypotheses 8 and 9 were not confirmed by the data. However, longitudinal observations might yield a deeper understanding in that respect. Effects that external stakeholders (e.g. customers and suppliers) have on the Fp of a company might be influenced by certain amount of time lag.

This has important implications for managers striving for success. There is a substantial consensus

today that a key competitive advantage of organizations lies in their ability to learn and to be responsive to challenges from both internal and external business environments. Clearly, more attention has to be paid to developing an organizational learning culture in order to improve organizational performance. This can be achieved by cultivating an environment in which the employees can and should continually learn and share their knowledge. One practical implication of this thinking is that investing effort, time and money into initiatives aimed at developing a learning-oriented culture can bring about improved performance—in terms of better relationships within and outside the company as well as in hard figures.

Another relatively straightforward implication of these research findings is that employees are the most important stakeholder group in any business process change initiative that is striving for improved organizational performance. Disregarding whether the approach to change is evolutionary (TQM) or revolutionary (BPR), employees are the focal group for management's attention. Hence, managers striving for effectiveness and efficiency in their processes should put people first.

Turning to the assumptions of the competing values model (Denison and Spreitzer, 1991), our

research confirms that, in reality, firms are combinations of all four ideal types of cultures. It is true that in an organizational learning culture group and developmental cultures are predominant. Still, hierarchical and rational cultures are present to some extent. While a flexibility orientation is considered as offering responses to most challenges in the modern business environment this is not necessarily always so. Besides innovation and creativity and openness and commitment, firms need some structure, stability and continuity while not neglecting the fact that they also have to be accomplishment-oriented.

The cross-sectional nature of the data gathered imposes the first methodological limitation—the inability to directly draw conclusions through a causal inference. Even though the structural equation models are conceptualized in a causal way, this technique needs to be backed up in advance with theoretical assumptions and previous research findings. For instance, in our case it might also be the case that superior organizational performance allows for the attainment of a higher-level organizational learning culture. For this reason, other research designs such as experimental and longitudinal ones are desirable when examining causal relationships among organizational variables (Egan et al., 2004). Such designs are extremely rare in this research context, whereas problems with data gathering (e.g. small overlaps of samples among different years, respondent non-disclosure, etc) occur on a regular basis. Nevertheless, Hult et al. (2002) did one similar study by conducting a longitudinal study of the learning climate on cycle times in supply chains.

Every researcher and manager dealing with organizational culture and business process change needs to be aware of the multiplexity and multiple dimensions of organizational culture (Schein, 1992; Trompenaars and Woolliams, 2003). Further, the existence of different kinds of subcultures within organizations also needs to be accounted for. Besides, organizational culture is also heavily intertwined with national culture and other contextual variables (Hofstede, 1980), which will all need to be considered in future research. Case studies should also be performed to further validate our findings. This would yield a deeper understanding of the relationship of the constructs proposed in our model and serve to further test its applicability and usefulness from the practical point of view.

Appendix A. List of measurement items

Information acquisition	
IA1	Employees in our organization are an extremely important source of information.
IA2	Previous decisions are a very useful source of information for current decisions.
IA3	New business methods and services are always worth trying even if they may prove risky.
IA4	Reports prepared by external experts are an extremely important source of information.
IA5	Our organization uses a dipping service — regular collection of papers and articles to our interest.
IA6	Our competitors are an extremely important source for learning new methods and services.
IA7	Expertise on the industry, products, and services is an extremely important criterion for hiring a new employee.
IA8	Joint tasks and mergers contribute a great deal of knowledge about industry and economic environment, new methods and services/products.
IA9	Top managers in any important decision seek information or advice from the board of directors or owners (in general).
IA10	Top managers in any important decision seek information or advice from source outside the company (hiring experts, contacting top managers of other companies, etc.).
IA11	Our organization has employees whose job is related to searching for external information.
IA12	External sources (reports, consultants, newsletters, etc.) are extremely important for the operation of our organization.
IA13	In our organization we explicitly reward employees who are a source of quality information.
Information interpretation	
II1	Personal contacts
II2	Team meetings
II3	Committees as decision-makers
II4	Telephone contacts

II5	Written memos, notes, letters		SUPPLIER	Relations with suppliers are very poor and unstable and usually short-term—Statement A
II6	Special reports			
II7	Formal chain of command reporting			
II8	Company's intranet as a means of interpreting information			We consider our relations with suppliers to be excellent because we maintain genuine partnerships with them—Statement B
II9	Forums (e-chat, e-debates)			
II10	The more information the subordinate has the better he/she will perform.		EMP1	The net fluctuation of employees (number of staff replaced due to dissatisfaction with pay, relationships in the workplace and chances for career advances, etc—Memo! reasons) is very high within our company—Statement A
II11	Information to a subordinate must always be simple and concise.			There are no cases in our company of people leaving for internal reasons—Statement B
Behavioral and cognitive changes				
BCC1	Adaptability to environmental pressures			
BCC2	Quality of products/services			
BCC3	Number of products/services offered		EMP2	Productivity of employees is much lower than the industry standard—Statement A
BCC4	Technology of operation			
BCC5	Speed of operations			
BCC6	Introduction of new marketing approaches			Productivity of employees is much higher than the industry standard—Statement B
BCC7	Average productivity of employees			
BCC8	Satisfaction of employees		EMP3	Employees' trust in leadership is low—Statement A
BCC9	Overall atmosphere			
BCC10	Personal communication between top managers and employees			Employees' trust in leadership is high—Statement B
BCC11	Team meetings' efficiency		EMP4	Trust among employees themselves is weak—Statement A
BCC12	Employees' level of understanding of the company's strategic orientation			Trust among employees is strong—Statement B
BCC13	Employee's level of understanding of major problems in the company		EMP5	Work organization is inefficient—Statement A
BCC14	Efficiency of information systems within the Company			Work organization is efficient—Statement B
Organizational performance				
ROA	Return on assets (ROA, %) in our company is well below the industry average—Statement A			Employees do not feel special commitment to the organization—Statement A
	Return on assets (ROA, %) in our company is well above the industry average—Statement B		EMP7	Employees feel very committed to the organization—Statement B
VAEMP	Value added per employee in our company is well below the industry average—Statement A			Employees are not prepared to go the extra mile for the company—Statement A
	Value added per employee in our company is well above the industry average—Statement B		EMP8	Employees are prepared to go the extra mile for the company—Statement B
				Work costs per employee are well above the industry average—Statement A
				Work costs per employee are well below the industry average—Statement B

EMP9	Absenteeism in our company is (relative to the competition) very high—Statement A Absenteeism in our company is (relative to the competition) very low—Statement B
EMP10	Employees are very dissatisfied with the situation within the company—Statement A Employees are very satisfied with the situation within the company—Statement B
EMP11	Learning ability and adaptability of employees is low (in comparison to the competition)—Statement A Learning ability and adaptability of employees is high (in comparison to the competition)—Statement B
EMP12	Risk-taking within the company is worse than it is with our competitors—Statement A Risk-taking within the company is better than it is with our competitors—Statement B
CUST1	The number of customer complaints within the last period has increased strongly—Statement A The number of customer complaints within the last period has decreased strongly—Statement B
CUST2	Speed of dealing with customer complaints (comparative to the competition] is low—Statement A We deal with customer complaints faster than our competition—Statement B
CUST3	We lose existing clients and do not manage to attract new ones—Statement A We retain existing clients and managed to attract new ones—Statement B
CUST4	Reputation of our company in eyes of the customers has declined—Statement A Reputation of our company in eyes of the customers has improved—Statement B

Appendix B. Results of exploratory factor analysis**INFORMATION ACQUISITION**

Varimax-rotated factor loadings

	Factor 1	Factor 2
IA1	0.225	0.290
IA2	0.022	0.247
IA3	0.123	0.175
IA4	0.168	0.477
IA5	0.414	0.262
IA6	0.197	0.462
IA7	0.038	0.327
IA8	0.141	0.468
IA9	0.083	0.280
IA10	0.256	0.442
IA11	0.775	−0.064
IA12	0.385	0.541
IA13	0.424	0.268

Factor 1: 5,11,13—clippings, have employees who search for information, reward employees' quality information—**OPERATIONAL AND TACTICAL INFORMATION ACQUISITION**

Factor 2: 4,6,8,10,12—external experts, competitors, joint tasks and mergers, top managers seek external information, external sources important for the operations—**STRATEGIC IA Out: 1,2,3,7,9**

Please note: after CFA item IA8 was then omitted from further analysis

INFORMATION INTERPRETATION

	Factor 1	Factor 2	Factor 3
II1	0.092	0.349	0.088
II2	0.088	0.850	0.180
II3	0.219	0.624	0.179
II4	0.149	0.094	0.385
II5	0.125	0.142	0.680
II6	0.102	0.160	0.691
II7	0.024	0.047	0.564
II8	0.490	0.146	0.297
II9	0.986	0.055	0.138
II10	0.040	0.339	0.020
II11	−0.114	0.271	0.042

Factor 1: E-interpretation—8,9 (intranet, e-forums)
Factor 2: Face-to-face interpretation—2,3 (team meetings, committees as decision-makers)

Factor 3: Formalized (impersonal) interpretation—
5,6,7 (written memos, notes, letters /special reports/
formal chain of reporting

Out: 1, 10,11

BEHAVIORAL AND COGNITIVE CHANGES

	Factor 1	Factor 2
BCC1	0.114	0.455
BCC2	0.233	0.532
BCC3	0.048	0.461
BCC4	0.275	0.686
BCC5	0.210	0.626
BCC6	0.186	0.492
BCC7	0.355	0.436
BCC8	0.844	0.197
BCC9	0.882	0.162
BCC10	0.664	0.172
BCC11	0.415	0.432
BCC12	0.534	0.405
BCC13	0.505	0.344
BCC14	0.392	0.534

Behavioral changes—Factor 1: 1–7

Cognitive changes—Factor 2: 8–10, 13

Out: 11,12,14

Please note: After CFA—items BCC1, BCC3,
BCC6 were additionally omitted

EMPLOYEE PERSPECTIVE OF PERFORMANCE—first iteration

	Factor 1	Factor 2
Emp1	0.424	0.205
Emp2	0.325	0.665
Emp3	0.667	0.382
Emp4	0.602	0.392
Emp5	0.661	0.395
Emp6	0.871	0.177
Emp7	0.776	0.345
Emp8	0.150	0.526
Emp9	0.217	0.486
Emp10	0.645	0.404
Emp11	0.468	0.448
Emp12	0.304	0.503

EMP SUBJECTIVE ('soft') measures: Factor 1:
3,6,7: trust in leadership, commitment, extra mile
for the company.

EMP OBJECTIVE ('hard') measures: Factor 2:
2,8,9: productivity of employees, work costs per
employee, absenteeism.

Out: 1, 10,11.

Please note: after several iterations other items
were excluded as well: 4,5,12.

CUSTOMER PERSPECTIVE OF PERFORMANCE

	Factor 1
CUST1	0.683
CUST2	0.776
CUST3	0.755
CUST4	0.812

One factor—all four variables.

Please note: after CFA—CUST1 is excluded as
well

ROA in VAEMP—each one is measurement
variable (composed out of 1 item)

SUP = 1 item

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